

Learning PGC

1 Representation of the Ternary Mixture Model

We consider a Probabilistic Generating Circuit (PGC) modeling a mixture of r product distributions over m ternary variables. For a single ternary variable x_j , the local marginal distribution can be parameterized by a quadratic polynomial of the form $(a_{ij} + b_{ij}x_j + c_{ij}x_j^2)$, where the coefficients correspond to the unnormalized probabilities or feature mappings of the three states.

The generating polynomial for the entire mixture model is given by:

$$f(x_1, x_2, \dots, x_m) = \sum_{i=1}^r \mu_i \prod_{j=1}^m (a_{ij} + b_{ij}x_j + c_{ij}x_j^2) \quad (1)$$

where μ_i represents the mixture weight (prior probability) of the i -th component.

1.1 Implicit Parameter Reduction

A general ternary distribution over a single variable has 3 states, but because these states must satisfy an affine normalization constraint (e.g., summing to 1), the distribution only possesses 2 degrees of freedom. We observe that:

$$a_{ij} + b_{ij} + c_{ij} = 1 \quad (2)$$

Because of this linear dependence, the third parameter c_{ij} is entirely implicit ($c_{ij} = 1 - a_{ij} - b_{ij}$). Therefore, we can drop c_{ij} and fully encode the state of the j -th variable in the i -th bag using a 2-dimensional vector: $[a_{ij}, b_{ij}]^T$.

By concatenating these localized parameters across all m variables, we define the global parameter vector $\alpha_i \in \mathbb{R}^{2m}$ for the i -th mixture component:

$$\alpha_i = [a_{i,1}, b_{i,1}, a_{i,2}, b_{i,2}, \dots, a_{i,m}, b_{i,m}]^T \quad (3)$$

The complete parameter matrix for the model is $A \in \mathbb{R}^{2m \times r}$, where $A = [\alpha_1, \alpha_2, \dots, \alpha_r]$.

2 Extraction of Second and Third-Order Coefficients

To perform tensor decomposition, we must construct the second-order (M_2) and third-order (M_3) observable moments. We extract the pairwise and triple-wise interactions of our parameters by marginalizing out irrelevant variables and applying discrete derivatives.

To isolate the interaction between two distinct variables J and K ($J \neq K$), we evaluate the generating polynomial by setting $x_s = 1$ for all $s \notin \{J, K\}$. Let this marginalized bivariate polynomial be $f_{J,K}(x_J, x_K)$.

- By evaluating $f_{J,K}$ at specific target points $\{-1, 0, 1\}$, we can extract the exact bilinear coefficients corresponding to the components of α_i :

- **Extracting the $a \cdot a$ cross-terms:** Evaluating at the origin eliminates the linear and quadratic terms:

$$f_{J,K}(0,0) = \sum_{i=1}^r \mu_i a_{iJ} a_{iK} \quad (4)$$

- **Extracting the mixed $a \cdot b$ and $b \cdot a$ cross-terms:** We isolate the linear x_K coefficient by taking a finite difference with respect to x_K , while holding $x_J = 0$:

$$\frac{f_{J,K}(0,1) - f_{J,K}(0,-1)}{2} = \sum_{i=1}^r \mu_i a_{iJ} b_{iK} \quad (5)$$

By symmetry, holding $x_K = 0$ and differencing across x_J yields:

$$\frac{f_{J,K}(1,0) - f_{J,K}(-1,0)}{2} = \sum_{i=1}^r \mu_i b_{iJ} a_{iK} \quad (6)$$

- **Extracting the $b \cdot b$ cross-terms:** To extract the coefficient of the bilinear $x_J x_K$ term, we do the following :

$$\frac{f_{J,K}(1,1) - f_{J,K}(1,-1) - f_{J,K}(-1,1) + f_{J,K}(-1,-1)}{4} = \sum_{i=1}^r \mu_i b_{iJ} b_{iK} \quad (7)$$

By systematically computing these evaluations for all pairs $J \neq K$, we successfully reconstruct all cross-variable entries.

To isolate the third-order interaction between three distinct variables J, K , and L , we evaluate the generating polynomial by setting $x_s = 1$ for all $s \notin \{J, K, L\}$. Let this marginalized trivariate polynomial be $f_{J,K,L}(x_J, x_K, x_L)$.

By applying the same discrete operators—evaluating at 0 to extract the a parameter, and computing the central difference across $\{-1, 1\}$ to cleanly eliminate the quadratic c terms and extract the b parameter—we can reconstruct all third-order cross-terms for α_i :

- **Extracting the $a \cdot a \cdot a$ cross-terms:** Evaluating all three variables at the origin eliminates all linear and quadratic terms:

$$f_{J,K,L}(0,0,0) = \sum_{i=1}^r \mu_i a_{iJ} a_{iK} a_{iL} \quad (8)$$

- **Extracting the mixed terms with one b (e.g., $b \cdot a \cdot a$):** We isolate the linear x_J coefficient by taking a finite difference with respect to x_J , while holding $x_K = 0$ and $x_L = 0$:

$$\frac{f_{J,K,L}(1,0,0) - f_{J,K,L}(-1,0,0)}{2} = \sum_{i=1}^r \mu_i b_{iJ} a_{iK} a_{iL} \quad (9)$$

By symmetry, applying the difference operator strictly across x_K or x_L while evaluating the others at 0 yields the $a \cdot b \cdot a$ and $a \cdot a \cdot b$ coefficients, respectively.

- **Extracting the mixed terms with two b 's (e.g., $b \cdot b \cdot a$):** We apply a 2D discrete derivative mask across x_J and x_K , while evaluating $x_L = 0$:

$$\frac{f_{J,K,L}(1,1,0) - f_{J,K,L}(1,-1,0) - f_{J,K,L}(-1,1,0) + f_{J,K,L}(-1,-1,0)}{4} = \sum_{i=1}^r \mu_i b_{iJ} b_{iK} a_{iL} \quad (10)$$

This formulation extends symmetrically to extract the $b \cdot a \cdot b$ and $a \cdot b \cdot b$ coefficients.

- **Extracting the $b \cdot b \cdot b$ cross-terms:** To extract the pure trilinear $x_J x_K x_L$ coefficient, we apply a 3D discrete derivative mask across all three variables.

$$\begin{aligned} & \frac{1}{8} \left[f_{J,K,L}(1,1,1) - f_{J,K,L}(-1,1,1) - f_{J,K,L}(1,-1,1) + f_{J,K,L}(-1,-1,1) \right. \\ & \quad \left. - f_{J,K,L}(1,1,-1) + f_{J,K,L}(-1,1,-1) + f_{J,K,L}(1,-1,-1) - f_{J,K,L}(-1,-1,-1) \right] \\ & = \sum_{i=1}^r \mu_i b_{iJ} b_{iK} b_{iL} \end{aligned} \quad (11)$$

By systematically iterating these operations over all triplets $J < K < L$, we fully populate the off-diagonal components of the third-order tensor M_3 .

3 Small restriction

After preprocessing the input, we may assume w.l.o.g. that the generating polynomial f has no nontrivial common factor and that every variable appears in f .

To simplify the parametrization for the ternary case, we normalize the local polynomials. Let the local marginal for variable x_j in component i be given by $a_{i,j} + b_{i,j}x_j + c_{i,j}x_j^2$. Assuming $a_{i,j} \neq 0$, we can factor it out (absorbing it into the mixture weight μ_i) to rewrite the polynomial factor as $1 + \tilde{a}_{i,j}x_j + \tilde{b}_{i,j}x_j^2$, where:

$$\tilde{a}_{i,j} = \frac{b_{i,j}}{a_{i,j}} \quad \text{and} \quad \tilde{b}_{i,j} = \frac{c_{i,j}}{a_{i,j}} \quad (12)$$

Next choose an *alive set* $A \subseteq [n]$ of size m , and let $F := [n] \setminus A$ be the set of *fixed variables*. Permuting indices if needed, assume $A = \{1, 2, \dots, m\}$. We set the fixed variables to 0, i.e., $x_F = \mathbf{0}$, which evaluates their respective polynomial factors to 1. We obtain an m -variate restricted polynomial $h(x_1, \dots, x_m)$:

$$h(x_1, \dots, x_m) = \sum_{i=1}^r \mu_i \prod_{j=1}^m \left(1 + \tilde{a}_{i,j}x_j + \tilde{b}_{i,j}x_j^2 \right) \quad (13)$$

Once we have h , we compute the symbolic rank of the Jacobian to argue about the preservation of mixture size.

Intuition. Fix an alive set A of size m and view the coefficient vector $C = (c_E)_{E \in \{0,1,2\}^m}$ of h as a function of the hidden parameters $V := \{\mu_i\}_{i \in [r]} \cup \{\tilde{a}_{i,j}, \tilde{b}_{i,j}\}_{i \in [r], j \in [m]}$. This defines a polynomial map from parameters to observable coefficients. An r -mixture has more degrees of freedom than an $(r-1)$ -mixture, so generically its image has a larger local dimension. We certify this separation by lower-bounding the symbolic rank of the Jacobian.

For any degree tuple $E \in \{0, 1, 2\}^m$, the coefficient of the monomial $\prod x_j^{E_j}$ is given by:

$$c_E = \sum_{i=1}^r \mu_i \prod_{j:E_j=1} \tilde{a}_{i,j} \prod_{j:E_j=2} \tilde{b}_{i,j} \quad (14)$$

Let $C = (c_E) \in \mathbb{R}^{3^m}$ denote the vector of all coefficients. For each $i \in [r]$ and $E \in \{0, 1, 2\}^m$, the partial derivatives are:

$$\frac{\partial c_E}{\partial \mu_i} = \prod_{E_j=1} \tilde{a}_{i,j} \prod_{E_j=2} \tilde{b}_{i,j} \quad (15)$$

$$\frac{\partial c_E}{\partial \tilde{a}_{i,l}} = \mu_i \cdot \mathbf{1}_{\{E_l=1\}} \prod_{\substack{j \neq l \\ E_j=1}} \tilde{a}_{i,j} \prod_{E_j=2} \tilde{b}_{i,j} \quad (16)$$

$$\frac{\partial c_E}{\partial \tilde{b}_{i,l}} = \mu_i \cdot \mathbf{1}_{\{E_l=2\}} \prod_{E_j=1} \tilde{a}_{i,j} \prod_{\substack{j \neq l \\ E_j=2}} \tilde{b}_{i,j} \quad (17)$$

Viewing the coefficients $\{c_E\}$ as polynomials in the parameter set V , we form the Jacobian matrix J_r with 3^m rows indexed by $E \in \{0, 1, 2\}^m$ and $r(2m+1)$ columns indexed by variables $v \in V$, with entries $J_{E,v} := \frac{\partial c_E}{\partial v}$.

Let M_r denote the submatrix of J_r obtained by restricting to the columns indexed by $\{\tilde{a}_{i,l}, \tilde{b}_{i,l}\}$ and to the rows indexed by $E \neq \mathbf{0}$. Thus M_r has size $(3^m - 1) \times (2mr)$. In J_r , the μ -columns contribute at most r additional rank, while the row $E = \mathbf{0}$ contributes one to the rank (because $\partial c_0 / \partial \mu_i = 1$ for all i and $\partial c_0 / \partial \text{params} = 0$). Therefore,

$$1 + \text{rank}(M_r) \leq \text{rank}(J_r) \leq r + \text{rank}(M_r) \quad (18)$$

Hence, it suffices to determine $\text{rank}(M_r)$. In fact, we can show that $\text{rank}(M_r) = 2mr$ in the generic setting. Thus, we get:

$$2mr + 1 \leq \text{rank}(J_r) \leq 2mr + r \quad (19)$$

Separating r from $r-1$. Once we have the above relation, we can separate J_r from J_{r-1} using the rank bounds. For an $(r-1)$ -mixture, applying the upper bound yields:

$$\text{rank}(J_{r-1}) \leq (r-1) + \text{rank}(M_{r-1}) \leq (r-1) + 2m(r-1) \quad (20)$$

since M_{r-1} has at most $2m(r-1)$ columns. We require the minimum rank of the r -mixture to strictly exceed the maximum rank of the $(r-1)$ -mixture:

$$2mr + 1 > (r-1) + 2m(r-1) \quad (21)$$

$$2mr + 1 > 2mr - 2m + r - 1 \quad (22)$$

$$2m > r - 2 \quad (23)$$

Because m and r are integers, this simplifies to $2m \geq r - 1$. Under this condition, the Jacobian ranks of the r - and $(r-1)$ -mixture models are separated. This leads us to the following proposition:

Proposition 3.1. *For $2m \geq r - 1$, in the generic setting, restricting to any alive set of size m does not reduce the top fan-in of the ternary PGC, i.e., the mixture size remains r .*

3.1 Rank bound on the Jacobian

In this section, we prove the rank-separation for the general r -component ternary mixture. Recall that the Jacobian J_r has rows indexed by the degree tuples $E \in \{0, 1, 2\}^m$ and columns indexed by the parameters $\{\mu_i\}_{i \in [r]} \cup \{\tilde{a}_{i,l}, \tilde{b}_{i,l}\}_{i \in [r], l \in [m]}$. Let M_r denote the submatrix of J_r obtained by restricting to the columns indexed by $\{\tilde{a}_{i,l}, \tilde{b}_{i,l}\}$ and rows indexed by $E \neq \mathbf{0}$. Thus M_r has size $(3^m - 1) \times 2mr$. We will show that generically, $\text{rank}(M_r) = 2mr$.

To lower bound the symbolic rank of M_r , it is enough to exhibit one assignment of parameters for which M_r attains large rank. We specialize to:

$$\tilde{a}_{i,j} = a_i \cdot u_j \quad \text{and} \quad \tilde{b}_{i,j} = c_i \cdot v_j \quad \text{for all } i \in [r], j \in [m], \quad (24)$$

where $a_i, c_i, u_j, v_j \neq 0$. (Since we only use this as a witness assignment for rank, the signs are irrelevant).

To construct a full-rank square submatrix, we strategically isolate the parameters by restricting our attention to “pure” degree tuples. Let $\mathcal{E}_k^{(1)} = \{E \in \{0, 1, 2\}^m : |\{j : E_j = 1\}| = k, |\{j : E_j = 2\}| = 0\}$ be the set of tuples containing exactly k ones and no twos. Symmetrically, let $\mathcal{E}_k^{(2)}$ be the set of tuples containing exactly k twos and no ones.

For any $E \in \mathcal{E}_k^{(1)}$, the indicator $\mathbf{1}_{\{E_\ell=2\}} = 0$ for all ℓ . Therefore, all partial derivatives with respect to $\tilde{b}_{i,\ell}$ vanish identically. The remaining non-zero derivatives simplify to:

$$\frac{\partial c_E}{\partial \tilde{a}_{i,\ell}} = \mu_i a_i^{k-1} \cdot \mathbf{1}_{\{E_\ell=1\}} \prod_{\substack{j \neq \ell \\ E_j=1}} u_j \quad (25)$$

Notice that this derivative structure is exactly identical to the binary setting. Thus, grouping the rows of M_r indexed by $\mathcal{E}_k^{(1)}$, we obtain a block-row of the form:

$$M_{\mathcal{E}_k^{(1)}} = \left(\mu_1 a_1^{k-1} G_k \quad \mu_2 a_2^{k-1} G_k \quad \dots \quad \mu_r a_r^{k-1} G_k \quad \middle| \quad \mathbf{0} \quad \mathbf{0} \quad \dots \quad \mathbf{0} \right) \quad (26)$$

where G_k is the standard $\binom{m}{k} \times m$ incidence matrix scaled by $(u_1, \dots, u_m)$ exactly as defined in the binary model.

Symmetrically, for $E \in \mathcal{E}_k^{(2)}$, all derivatives with respect to $\tilde{a}_{i,\ell}$ vanish, yielding the block-row:

$$M_{\mathcal{E}_k^{(2)}} = \left(\mathbf{0} \quad \mathbf{0} \quad \dots \quad \mathbf{0} \quad \middle| \quad \mu_1 c_1^{k-1} G_k \quad \mu_2 c_2^{k-1} G_k \quad \dots \quad \mu_r c_r^{k-1} G_k \right) \quad (27)$$

where G_k is identically structured but scaled by the parameters $(v_1, \dots, v_m)$.

Reducing to a Vandermonde structure. By Lemma C.2 of the binary setting, for each $k \in \{1, \dots, m-1\}$, the generic incidence matrix G_k has full column rank m . Therefore, there exist invertible matrices of row operations R_k such that $R_k G_k$ has the form $\begin{bmatrix} I_m \\ \mathbf{0} \end{bmatrix}$.

Applying these row operations independently to the pure blocks $M_{\mathcal{E}_k^{(1)}}$ and $M_{\mathcal{E}_k^{(2)}}$ (and discarding all-zero rows), we obtain a reduced submatrix M' of M_r :

$$M'_{\mathcal{E}_k^{(1)}} = \left(\mu_1 a_1^{k-1} I_m \quad \dots \quad \mu_r a_r^{k-1} I_m \quad \middle| \quad \mathbf{0} \quad \dots \quad \mathbf{0} \right) \quad (28)$$

$$M'_{\mathcal{E}_k^{(2)}} = \left(\mathbf{0} \quad \dots \quad \mathbf{0} \quad \middle| \quad \mu_1 c_1^{k-1} I_m \quad \dots \quad \mu_r c_r^{k-1} I_m \right) \quad (29)$$

Clearly $\text{rank}(M_r) = \text{rank}(M')$.

Now assume $m-1 \geq r+1$. Choose $\mu_i \neq 0$, pairwise distinct $a_1, \dots, a_r$, and pairwise distinct $c_1, \dots, c_r$. We construct a square $2mr \times 2mr$ test matrix V by extracting the first r block-rows of type $\mathcal{E}^{(1)}$ (i.e., $k = 1, \dots, r$) and the first r block-rows of type $\mathcal{E}^{(2)}$ (i.e., $k = 1, \dots, r$).

Because the blocks are strictly decoupled, the resulting submatrix V is block-diagonal:

$$V = \begin{pmatrix} (A_{r \times r} \otimes I_m) \cdot D_\mu & \mathbf{0} \\ \mathbf{0} & (C_{r \times r} \otimes I_m) \cdot D_\mu \end{pmatrix} \quad (30)$$

where $D_\mu = \text{diag}(\mu_1, \dots, \mu_r) \otimes I_m$, and $A_{r \times r}$ and $C_{r \times r}$ are standard Vandermonde matrices generated by a_i and c_i respectively.

The determinant of a block-diagonal matrix is the product of the determinants of its diagonal blocks. Since $A_{r \times r}$ and $C_{r \times r}$ are non-singular Vandermonde matrices, both diagonal blocks have full rank mr . Therefore, $\text{rank}(V) = mr + mr = 2mr$.

Because V is a submatrix of M_r , we have $\text{rank}(M_r) \geq 2mr$. Since M_r possesses exactly $2mr$ columns, $\text{rank}(M_r) = 2mr$ generically. Consequently, the symbolic rank of M_r is $2mr$ whenever $m \geq r + 1$.

Combining with the boundaries, for $m \geq r + 1$ we obtain:

$$2mr + 1 \leq \text{rank}(J_r) \leq 2mr + r \tag{31}$$